Notes on Spacetime and Geometry: An Introduction to General Relativity

Wang Zhen
2150946@tongji.edu.cn

School of Physics Science and Engineering, Tongji University, 1239 Siping Road, 200433 Shanghai, China

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Contents

Abstract

These notes are based on the book *Spacetime and Geometry: An Introduction to General Relativity* by Sean Carroll. The primary objective is to aid in the personal comprehension and formulation of the material presented in the book, rather than to generate profit or serve commercial purposes. Given the limitations inherent in this process, occasional typographical errors may occur. I use ChatGPT to improve the text.

1 Special Relativity and Flat Spacetime

1.1 Prelude

- General relativity (GR) is Einstein's theory of space, time and gravitation. At heart it is a very simple subject (compared, for example, to anything involving quantum mechanics).
- Our task is to understand spacetime, curvature, and how curvature becomes gravity.

1.2 Space and Time, Separately and Together

- **Spacetime** is a four-dimensional set, with elements labeled by three dimensions of space and one of time. An individual point in spacetime is called an **event**. The path of a particlee is a curve through spacetime, a parametrized one-dimensional set of events, called **worldline**.
- **Spacetime interval** between two points is defined as

$$(\Delta s)^2 = -(c\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2. \tag{1.1}$$

It can be positive, negative or zero. The point is that it is invariant under changes of inertial coordinates. In a more compact form, we introduce the **metric**:

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{1.2}$$

- The **proper time**

$$(\Delta \tau)^2 = -(\Delta s)^2 = -\eta_{\mu\nu}\Delta x^\mu \Delta x^\nu \tag{1.3}$$

between two events measures the time elapsed as seen by an observer moving on a straight path between the events. For general trajectories, the proper time and coordinate time are different.

1.3 Lorentz Transformations

- The matrices that satisfy

$$\eta = \Lambda^{\mathrm{T}} \eta \Lambda \tag{1.4}$$

are known as the **Lorentz transformations**. The set of them forms a group under matrix multiplication, known as the **Lorentz group**. To be precise, the elements of the group are three **rotations** and three **boosts**.
- Rotations, boosts and **translations** constitute the **Poincaré group**, which is a ten-parameter group.

1.4 Vectors

- Of course, in spacetime vectors are four-dimensional, and are often referred to as **four-vectors**. This turns out to make quite a bit of difference. For example, there is no such thing as a cross product between two four-vectors.
- to each point p in spacetime we associate the set of all possible vectors located at that point; this set is known as the tangent space at p, or T_p.

- A (real) vector space is a collection of objects (vectors) that can be added together and multiplied by real numbers in a linear way.
- A vector is a perfectly well-defined geometric object, as is a vector field, defined as a set of vectors with exactly one at each point in spacetime.
- Under Lorentz transformations, the components of a tangent vector must change as

$$V^\mu \to V^{\mu'} = \Lambda^{\mu'}{}_\nu V^\nu. \tag{1.5}$$

  The vector V itself does not change.

1.5 Dual Vectors

1.6 Tensors

1.7 Manipulating Tensors

- **Contraction** turns a (k, l) tensor into a $(k-1, l-1)$ tensor. Contraction proceeds by summing over one upper and one lower index.
- The metric and inverse metric can be used to **raise and lower indices** on tensors.
- We refer to a tensor as **symmetric** in any of its indices if it is unchanged under exchange of those indices. Thus if

$$S_{\mu\nu\rho} = S_{\nu\mu\rho}, \tag{1.6}$$

  we say that $S_{\mu\nu\rho}$ is symmetric in its first two indices. Similarly, a tensor is **antisymmetric** (or skew-symmetric) in any of its indices if it changes sign when those indices are exchanged; thus,

$$A_{\mu\nu\rho} = -A_{\rho\nu\mu} \tag{1.7}$$

  means that $A_{\mu\nu\rho}$ is antisymmetric in its first and third indices.

1.8 Maxwell's Equations

- **Maxwell equations** are bridges connecting electric field and magnetic field:

$$\begin{aligned} \nabla \times \vec{B} - \partial_t \vec{E} &= \vec{J} \\ \nabla \cdot \vec{E} &= \rho \\ \nabla \times \vec{E} + \partial_t \vec{B} &= 0 \\ \nabla \cdot \vec{E} &= 0. \end{aligned} \tag{1.8}$$

  With detailed calculations, we are able to know more about electromagnetic wave. These equations also provide great duality between electric and magnetic fields as shifting variables by $\vec{E} \to \vec{B}$ and $\vec{B} \to -\vec{E}$ does not change the equations.

- Under tensor forms, these can be expressed in more compact forms:

$$\partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0, \tag{1.9}$$

  where $F_{\mu\nu}$ is the field strength tensor.

1.9 Energy and Momentum

- The **momentum four-vector** is defined by

$$p^\mu = mU^\mu. \tag{1.10}$$

  The **energy** of a particle is simply the zeroth component of the momentum: $E = p^0$.

- A single momentum four-vector field is insufficient to describe the energy and momentum of a fluid; we must go further and define the **energy-momentum tensor** $T^{\mu\nu}$, which is a symmetric (2,0) tensor. To describe the general fluid in GR, we arrive at the concept of a **perfect fluid**. In this case, the energy-momentum tensor is given by

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix} \tag{1.11}$$

  A generalization to any frame gives

$$T^{\mu\nu} = (\rho + p)U^\mu U^\nu + p\eta^{\mu\nu}. \tag{1.12}$$

1.10 Classical Field Theory

- We can derive the equations of motion for such a particle by using the principle of least action: we search for critical points of an **action** S, written as

$$S = \int \mathrm{d}t\, L(q, \dot{q}), \tag{1.13}$$

  where the function $L(q, \dot{q})$ is the **Lagrangian**.

- Following the calculus-of-variations procedure, we show that critical points of action are those that satisfy the **Euler-Lagrangian equations**,

$$\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) = 0. \tag{1.14}$$

- In field theory, the Lagrangian an be expressed as an integral over space of a **Lagrangian density** $\mathscr{L}$:

$$L = \int \mathrm{d}^3x\, \mathscr{L}(\Phi_i, \partial_\mu \Phi_i). \tag{1.15}$$

- The **Klein-Gordon equation**

$$\Box\phi - m^2\phi = 0, \tag{1.16}$$

  where $\Box \equiv \eta^{\mu\nu}\partial_\mu\partial_\nu$ is known as the **d'Alembertian**.

2 Manifolds

2.1 Gravity as Geometry

- Weak equivalence principle (WEP): The motion of free-falling particles are the same in a gravitational field and a uniformly accelerated frame, in small enough regions of spacetime.

According to WEP, the inertial mass and gravitational mass of any object are equal:

$$m_i = m_g. \tag{2.1}$$

- Einstein equivalence principle (EEP): In small enough regions of spacetime, the laws of physics reduced to those of special relativity; it is impossible to detect the existence of gravitational field by means of local experiments.

A direct application of EEP is **gravitational redshift**, which is discussed in the context of two rockets with a uniform acceleration and a tower on the surface of a planet. This give rise to the fact that the time of the emission and reception of a photon at different elevation of the tower would not be the same.

To better describe such curved spacetime, we need the mathematical structure called **manifold**, which looks locally like Minkowski spacetime.

2.2 What Is a Manifold?

- Examples of manifolds: $\mathbb{R}^n$, n-sphere S^n, n-torus T^n, Lie group, ...
- A map ϕ can be classified as one-to-one (injective) and onto (surjective). Examples: $\phi(x) = e^x$ is one-to-one, but not onto; $\phi(x) = x^3 - x$ is onto, but not one-to-one; $\phi(x) = x^3$ is both; $\phi(x) = x^2$ is neither.

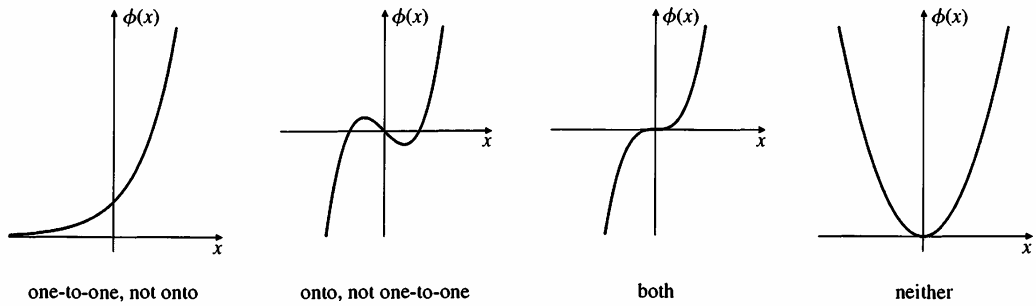

Figure 1: Types of maps.

2.3 Vectors Again

- Rigorously, the **tangent space** is not simply defined by all the **tangent vectors** at the point p but all the **vectors**. We used only things that are intrinsic to the manifold M, which means there are no embeddings in higher-dimensional spaces.

2.4 Tensors Again

2.5 The Metric

2.6 An Expanding Universe

- A simple example of a nontrivial Lorentzian geometry is provided by a four-dimensional cosmological spacetime with metric

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a^2(t)(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2). \tag{2.2}$$

  For a fixed moment of time, the universe described by this metric is a flat three-dimensional Euclidean space.

- Typical solutions for the scale factor are power laws

$$a(t) = t^q, \quad 0 < q < 1. \tag{2.3}$$

  A matter-dominated flat universe satisfies $q = 2/3$, while a radiation-dominated flat universe satisfies $q = 1/2$.

2.7 Causality

2.8 Tensor Densities

- Levi-Civita symbol $\tilde{\epsilon}_{\mu_1\mu_2\cdots\mu_n}$ is called a symbol because it is **not** a tensor and is defined invariant under coordinate transformations.

- It transforms in a way close to the tensor transformation law, except for the determinant out front:

$$\tilde{\epsilon}_{\mu_1'\mu_2'\cdots\mu_n'} = \left|\frac{\partial x^{\mu'}}{\partial x^{\mu}}\right| \tilde{\epsilon}_{\mu_1\mu_2\cdots\mu_n} \frac{\partial x^{\mu_1}}{\partial x^{\mu_1'}} \frac{\partial x^{\mu_2}}{\partial x^{\mu_2'}} \cdots \frac{\partial x^{\mu_n}}{\partial x^{\mu_n'}}. \tag{2.4}$$

  Objects transforming in this way are known as **tensor densities**.

- The power to which the Jacobian is raised is known as the **weight** of the tensor density. The Levi-Civita symbol is a density of weight 1, while g is a density of weight -2.

- We can define the **Levi-Civita tensor** as

$$\epsilon_{\mu_1\mu_2\cdots\mu_n} = \sqrt{|g|}\,\tilde{\epsilon}_{\mu_1\mu_2\cdots\mu_n}. \tag{2.5}$$

2.9 Differential Forms

- A differential p-form is simply a $(0, p)$ tensor that is completely symmetric.

- The **exterior derivative** d allows us to differentiate p-form fields to obtain $(p+1)$-form fields, satisfying the graded Leibniz rule:

$$\mathrm{d}(\omega \wedge \eta) = (\mathrm{d}\omega) \wedge \eta + (-1)^p \omega \wedge (\mathrm{d}\eta). \tag{2.6}$$

  The proof of equation (2.6) will be given in solutions to the exercises.

- A p-form is **closed** if $\mathrm{d}A = 0$, and **exact** if $\mathrm{d}A = B$ for some $(p-1)$-form B. All exact forms are closed but the converse is not necessarily true. The vector space consisting of elements called cohomology classes is defined as closed forms modulo the exact forms:

$$H^p(M) = \frac{Z^p(M)}{B^p(M)}. \tag{2.7}$$

It takes time but is worthwhile to cerebrate the fact that the information about topology can be extracted in terms of cohomology group.

- **Hodge dual** $(*)$ can be extremely useful in describing the electromagnetic field. It also provides us with profound insights on theories like QED and QCD, as they possess distinct coupling behaviors.

2.10 Integration

- On an n-dimensional manifold M, the integrand is properly understood as an n-form. In other words, an integral over an n-dimensional region $\Sigma \subset M$ is a map from an n-form field ω to the real numbers:

$$\int_\Sigma : \omega \to \mathbb{R}. \tag{2.8}$$

In one-dimensional case, you may be used to thinking of the symbol $\mathrm{d}x$ as an infinitesimal distance, but it is more properly a differential form. This intuitive understanding can be generalized into higher-dimensional case, as the wedge product of n copies of 1-form indeed assign a real number to every infinitesimal region, namely, the volume of that region.

- The integral of a scalar function ϕ over an n-manifold is written as

$$I = \int \phi(x) \sqrt{|g|}\, \mathrm{d}^n x\,. \tag{2.9}$$

It is clear that the volume element $\mathrm{d}^n x$ transforms as a density, instead of a tensor. This is the reason for multiplying an extra factor of $\sqrt{|g|}$ which helps construct an invariant volume element.

3 Curvature

3.1 Overview

We impose the most important definitions those will be used in this Chapter, without more detailed discussions here.

- The Christoffel symbol

$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} \left(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu} \right). \tag{3.1}$$

- The covariant derivative of a vector field is given by

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^{\lambda}_{\mu\nu} V^\nu. \tag{3.2}$$

- The geodesic equation

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\lambda^2} + \Gamma^{\mu}_{\rho\sigma} \frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda} \frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda} = 0. \tag{3.3}$$

  It is a curve parametrized by λ.

- The Riemann curvature tensor

$$R^{\rho}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda} \Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda} \Gamma^{\lambda}_{\mu\sigma}. \tag{3.4}$$

3.2 Covariant Derivatives

We would like to define a **covariant derivative** operator ∇ to perform the functions of partial derivative, but in a way independent of coordinates. From mathematical perspectives, it is required to satisfy these properties:

$$\text{linearity:} \quad \nabla(T + S) = \nabla T + \nabla S; \tag{3.5}$$

$$\text{Leibniz rule:} \quad \nabla(T \otimes S) = (\nabla T) \otimes S + T \otimes (\nabla S). \tag{3.6}$$

- We therefore have

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^{\nu}_{\mu\lambda} V^\lambda, \tag{3.7}$$

where matrices $(\Gamma_\mu)^\rho_\sigma$ are known as **connection coefficients**, satisfying equation (3.1). Just remember that they are **not** tensors.

- There is a tensor we can associate with any given connection known as the **torsion tensor**, defined by

$$T^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\mu\nu} - \Gamma^{\lambda}_{\nu\mu} = 2\Gamma_{[\mu\nu]}. \tag{3.8}$$

It is **torsion-free** if the right-hand side vanishes.

3.3 Parallel Transport and Geodesics

The result of parallel transporting a vector from one point to another will depend on the path taken between the points. Two vectors can only be compared in a natural way if they are elements of the same tangent space.

- For vectors, the equation of parallel transport takes the form

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} V^\mu + \Gamma^{\mu}_{\sigma\rho} \frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda} V^\rho = 0. \tag{3.9}$$

This can be viewed as a first-order differential equation with an initial value.

If the connection is metric-compatible, the metric is always parallel transported with respect to it:

$$\frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda}\nabla_\sigma g_{\mu\nu} = 0. \tag{3.10}$$

The parallel transport with respect to a metric-compatible connection preserves the norm of vectors, the sense of orthogonality, and so on.

- The geodesic equation is given by

$$\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda} = 0. \tag{3.11}$$

A better definition of a straight line is that it is a path that parallel-transports its own tangent vector. We can easily see that the equation (3.11) reproduces the usual notion of straight lines if the connection coefficients are the Christoffel symbols in Euclidean space.

3.4 Properties of Geodesics

- In terms of the four-momentum $p^\mu = mU^\mu$, the geodesic equation is simply

$$p^\lambda\nabla_\lambda p^\mu = 0. \tag{3.12}$$

This relation expresses the idea that freely-falling particles keep moving in the direction in which their momenta are pointing.

- Timelike geodesics maximizes proper time. This is how you can remember which twin in the twin paradox ages more–the one who stays home is basically on a geodesic, and therefore experiences more proper time.

3.5 The Expanding Universe Revisited

- Using variational method and comparing δI with geodesic equation, we are able to obtain the connection coefficients for the corresponding metric.
- The **cosmological redshift** is *not* a Doppler effect. A photon emitted with energy E_1 at scale factor a_1 and observed with energy E_2 at scale factor a_2 will have

$$\frac{E_2}{E_1} = \frac{a_1}{a_2}. \tag{3.13}$$

This is called a "redshift" because the wavelength of the photon is inversely proportional to the frequency. A greater redshift implies a greater distance. The amount of redshift is given by

$$z = \frac{\omega_1 - \omega_2}{\omega_2} = \frac{a_2}{a_1} - 1, \tag{3.14}$$

so that z vanishes if there has been no expansion.

- We typically model the matter filling the universe as a perfect fluid. A set of non-relativistic, non-interacting particles behaves like dust; a set of photons or other massless particles behaves like radiation, and a nonzero constant energy density throughout spacetime acts like vacuum.

3.6 The Riemann Curvature Tensor

There should be some tensor that tells us how the vector changes when it comes back to its starting point; it will be a linear transformation on a vector, and therefore involve one upper and one lower index. But it will also depend on the two vectors A and B that define the loop; therefore there should be two additional lower indices to contract with A and B.

- Thus the **Riemann curvature tensor** is defined as

$$R^{\rho}_{\sigma\mu\nu} = \partial_{\mu}\Gamma^{\rho}_{\nu\sigma} - \partial_{\nu}\Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}. \tag{3.15}$$

Note that this definition is completely constructed from connection that is a non-tensorial object.

- If a coordinate system exists in which the components of the metric are constant, the Riemann tensor will vanish.

3.7 Properties of the Riemann Tensor

- The Riemann tensor is antisymmetric in its first and last two indices:

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu}, \tag{3.16}$$

$$R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu}. \tag{3.17}$$

  It is invariant under interchange of the first pair of indices with the second:

$$R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}. \tag{3.18}$$

  With a little more work, we can see that the sum of cyclic permutations of the last three indices vanishes:

$$R_{\rho\sigma\mu\nu} + R_{\rho\nu\sigma\nu} + R_{\rho\mu\nu\sigma} = 0. \tag{3.19}$$

Our intuition is unfortunately contaminated by the fact that we are used to thinking about one- and two-dimensional spaces embedded in the (almost) Euclidean space in which we live, while the Riemann curvature is a property of the *intrinsic* geometry of a space, which could be measured by observers confined to the manifold.

- The **Ricci tensor** can be obtained by contracting the Riemann tensor:

$$R_{\mu\nu} = R^{\lambda}_{\mu\lambda\nu}, \tag{3.20}$$

  which is automatically symmetric.

- The trace of Ricci tensor is the **Ricci scalar**

$$R = R^{\mu}_{\mu} = g^{\mu\nu}R_{\mu\nu}. \tag{3.21}$$

- The Einstein tensor is defined as

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}. \tag{3.22}$$

3.8 Symmetries and Killing Vectors

- Symmetries of the metric are called **isometries**. Sometimes the existence of isometries is obvious, such as translations and Lorentz transformations for Minkowski spacetime.

- Any vector K_μ that satisfies $\nabla_\mu K_\nu + \nabla_\nu K_\mu = 0$ implies $K_\nu p^\nu$ is conserved along a geodesic trajectory. This equation is known as the **Killing equation**, and the vector fields that satisfy it are known as **Killing vectors**. (Recall equation (3.12)that the geodesic equation has something to do with the momentum p^μ.)
- Every Killing vector implies the existence of conserved quantities associated with geodesic motion.
- The linear combinations of Killing vector fields of constant coefficients is still a Killing vector field. The commutator of two Killing vector fields is a Killing vector field.

3.9 Maximally Symmetric Spaces

- The $\mathbb{R}^n$ space has
$$n + \frac{1}{2}n(n-1) = \frac{1}{2}n(n+1) \tag{3.23}$$
independent symmetries. You pick n axes to translate and two axes from n to rotate, and you got this result. Think about four-dimensional Minkowski spacetime, it has 10 symmetries (4 translations + 3 rotations + 3 boosts).
- If a manifold is maximally symmetric, the curvature is the same everywhere and the same in every direction. This is quite intuitive.
- In any maximally symmetric space, at any point, in any coordinate system, following equation holds:
$$R_{\rho\sigma\mu\nu} = \frac{R}{n(n-1)}\left(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}\right). \tag{3.24}$$

3.10 Geodesic Deviation

- The tangent vectors to the geodesics is defined as
$$T^\mu = \frac{\partial x^\mu}{\partial t}, \tag{3.25}$$
and the deviation vectors
$$S^\mu = \frac{\partial x^\mu}{\partial s}. \tag{3.26}$$
- The idea that S^μ points from one geodesic to the next inspire us to define the relative velocity of geodesics:
$$V^\mu = (\nabla_T S)^\mu = T^\rho \nabla_\rho S^\mu, \tag{3.27}$$
and the relative acceleration:
$$A^\mu = (\nabla_T V)^\mu = T^\rho \nabla_\rho V^\mu. \tag{3.28}$$
- The **geodesic deviation equation** is given by
$$A^\mu = \frac{D^2}{\mathrm{d}t^2}S^\mu = R^\mu_{\nu\rho\sigma}T^\nu T^\rho S^\sigma. \tag{3.29}$$

3.11 Selected Exercises Solutions

3. Imagine we have a diagonal metric $g_{\mu\nu}$. Show that the Chtistoffel symbols are given by

$$\Gamma^\lambda_{\mu\nu} = 0. \tag{3.30}$$

$$\Gamma^\lambda_{\mu\mu} = -\frac{1}{2}(g_{\lambda\lambda})^{-1}\partial_\lambda g_{\mu\mu}. \tag{3.31}$$

$$\Gamma^{\lambda}_{\mu\lambda} = \partial_{\mu}\left(\ln\sqrt{|g_{\lambda\lambda}|}\right). \tag{3.32}$$

$$\Gamma^{\lambda}_{\lambda\lambda} = \partial_{\lambda}\left(\ln\sqrt{|g_{\lambda\lambda}|}\right). \tag{3.33}$$

Solution:

Proof. We start with a known result from linear algebra:

$$\partial_{\mu}|g_{\lambda\lambda}| = |g_{\lambda\lambda}|g^{\lambda\sigma}\partial_{\mu}g_{\sigma\lambda}. \tag{3.34}$$

By definition (3.1), we expand the Christoffel symbol as:

$$\begin{aligned}\Gamma^{\lambda}_{\mu\lambda} &= \frac{1}{2}g^{\lambda\sigma}\left(\partial_{\mu}g_{\sigma\lambda} + \partial_{\lambda}g_{\mu\sigma} - \partial_{\sigma}g_{\mu\lambda}\right)\\ &= \frac{1}{2}g^{\lambda\sigma}\partial_{\mu}g_{\sigma\lambda}.\end{aligned} \tag{3.35}$$

Therefore, we find equation (3.31) could be proved:

$$\begin{aligned}\Gamma^{\lambda}_{\mu\lambda} &= \frac{1}{2}g^{\lambda\sigma}\partial_{\mu}g_{\sigma\lambda}\\ &= \frac{1}{2}\frac{1}{|g_{\lambda\lambda}|}\partial_{\mu}|g_{\lambda\lambda}|\\ &= \partial_{\mu}\left(\ln\sqrt{|g_{\lambda\lambda}|}\right).\end{aligned} \tag{3.36}$$

□

4 Gravitation

4.1 Physics in Curved Spacetime

- In general relativity, we would like to describe how the curvature of spacetime acts on matter to manifest itself as gravity, and how energy and momentum influence spacetime to create curvature.
- In its baldest form, the recipe of the **minimal-coupling principle** may be stated as follows
  (1) Take a law of physics that valid in inertial coordinates in flat spacetime.
  (2) Write it in a coordinate-invariant (tensorial) form.
  (3) Assert that the resulting law remains true in curved spacetime.

4.2 Einstein's Equation

We will make use of two ways to derive **Einstein's equation**. One way is to discuss on the plausible arguments, we have

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{4.1}$$

This tells us how the curvature of spacetime reacts to the presence of energy momentum. Einstein once thought that the left-hand side was nice and geometrical, while the right-hand side was somewhat less compelling.

- For the vacuum, it is obvious that $R_{\mu\nu} = 0$, which means nothing contributes to curvature.

4.3 Lagrangian Formulation

- Considering the action is a scalar, the simplest choice for a Lagrangian is

  $$S_H = \int \sqrt{-g}R \, \mathrm{d}^n x\,, \tag{4.2}$$

  known as the **Hilbert action** or sometimes the **Einstein-Hilbert action**. Please think of it carefully: why do we choose *Ricci scalar* to construct our Lagrangian? Why would minimal coupling make sense?

We can write variation of the action in a form of

$$\delta S_H = \delta S_1 + \delta S_2 + \delta S_3, \tag{4.3}$$

where

$$\begin{aligned} \delta S_1 &= \int \mathrm{d}^n x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu}, \\ \delta S_2 &= \int \mathrm{d}^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu}, \\ \delta S_3 &= \int \mathrm{d}^n x \, R \delta \sqrt{-g}. \end{aligned} \tag{4.4}$$

With more detailed calculations, we can also derive the Einstein's equation in the vacuum:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 0. \tag{4.5}$$

The advantage of the Lagrangian approach is manifested by the fact that our very first guess (which was practically unique) gave the right answer, in contrast with our previous trial-and-error method.

To get the non-vacuum field as well, we consider the action of the form

$$S = \frac{1}{16\pi G} S_H + S_M, \tag{4.6}$$

this will allow us to recover the equation (4.1).

- You must be proficient in manipulating tensors, and remember an important equation used in the above calculations:
$$\ln(\det M) = \mathrm{Tr}(\ln M). \tag{4.7}$$

4.4 Properties of Einstein's Equation

- The Einstein's equation is *nonlinear*, which is unique from other theories. For electromagnetism (or quantum electrodynamics), photon does not interact with themselves. However, graviton, if exists, interacts with all other particles, including gravitons.
- By investigating Raychaudhuri's equation, we get to know that gravity is attractive.
- At sufficiently high energy scales, GR may not be valid, thus we are looking for a better theory to describe quantum gravity. For example, the string theory.

4.5 The Cosmological Constant

- In order for static cosmology to solve the field equation with an ordinary matter source, it was necessary to add a new term called the **cosmological constant**, Λ, which enters as

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{4.8}$$

This is precisely equivalent to introducing a vacuum energy density

$$\rho_{\mathrm{vac}} = \frac{\Lambda}{8\pi G}. \tag{4.9}$$

The cosmological constant can be placed either at the left hand side or the right hand side of the equation. It serves as negative pressure, keeping the universe static.

4.6 Energy Conditions

- In principle, we can write down any energy-momentum tensor for Einstein equation, however, we should be responsible for the physical significance of the equation. In this sense, we introduce different energy conditions.

4.7 The Equivalence Principle Revisited

4.8 Alternative Theories

5 The Schwarzschild Solution

5.1 The Schwarzschild Metric

- The general solution to the Einstein's equation is the **Schwarzchild solution**:

$$ds^2 = -\left(1 - \frac{2GM}{R}\right) dt^2 + \left(1 - \frac{2GM}{R}\right)^{-1} dr^2 + r^2\, d\Omega^2\,, \tag{5.1}$$

which describes a spherically symmetric black hole that has neither electric charge nor angular momentum. Therefore, the **Schwarzschild radius** is given by:

$$r = 2GM, \tag{5.2}$$

which give rise to a coordinate singularity that can be wiped out by coordinate transformation, as we will see later. The point $r = 0$ is the authentic singularity.

- Note that as $M \to 0$ we recover the Minkowski space. This property is known as **asymptotic flatness**.

5.2 Birkhoff's Theorem

- **Birkhoff's theorem** is the statement that the Schwarzschild metric is the unique vacuum solution with spherical symmetry.

5.3 Singularities

- As mentioned, the Schwarschild metric has two singularities at $r = 0$ and $r = 2GM$, while the latter is not a physical singularity.
- To check singularities, we can construct tensors. For metric (5.1), direct calculation reveals that

$$R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma} = \frac{48G^2M^2}{r^6}. \tag{5.3}$$

This is enough to convince us that $r = 0$ represents an honest singularity.

5.4 Geodesics of Schwarzschild

- We should consider the behavior of geodesics to understand the Schwarzschild metric more in depth. It's easy to derive nonzero Christoffel symbols, and then work out geodesic equations.
- Symmetries simplify our calculations. We take timelike Killing vector

$$K_\mu = \left(-\left(1 - \frac{2GM}{r}\right), 0, 0, 0\right) \tag{5.4}$$

and

$$R_\mu = (0, 0, 0, r^2 \sin^2\theta). \tag{5.5}$$

- Conservation of angular momentum implies us that we can choose $\theta = \pi/2$, thus $\sin\theta = 1$.

The two conserved quantities are

$$E = -K_\mu \frac{dx^\mu}{d\lambda} = -\left(1 - \frac{2GM}{r}\right)\frac{dt}{d\lambda}, \tag{5.6}$$

and

$$L = R_\mu \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda} = r^2 \frac{\mathrm{d}\phi}{\mathrm{d}\lambda}. \tag{5.7}$$

For massless particles, these can be thought of as the conserved energy and angular momentum, while for massive particles they are the conserved energy and angular momentum per unit mass of the particle.

- The conserved quantities provide a convenient way to understand the orbits of particles in the Schwarzschild geometry. The potential for a particle can be obtained as

$$V(r) = \frac{1}{2}\epsilon - \epsilon \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \tag{5.8}$$

where the last term represents the GR contribution. For small r, it will turn out to make a great deal of difference.

5.5 Experimental Tests

- Originally Eistein suggested three tests to GR: **the deflection of light**, **the precession of perihelia**, and **gravtitational redshift**.
- The gravitational redshift was first detected in 1960 by Pound and Rebka, using gamma rays traveling upward a distance of only 72 feet. Subsequent tests have become increasingly precise. Further tests include binary pulsar, gravitational time delay, Big-Bang nucleosynthesis and so on, some among which will be discussed in later chapters.

5.6 Schwarzschild Black Holes

- The Schwarzschild metric:

$$\mathrm{d}s^2 = -\left(1 - \frac{2GM}{r}\right)\mathrm{d}t^2 + \left(1 - \frac{2GM}{r}\right)^{-1}\mathrm{d}r^2. \tag{5.9}$$

One way to understand the geometry of a spacetime is to explore its causal structure, as defined by the light cones. Letting the metric to be zero, we see that

$$\frac{\mathrm{d}t}{\mathrm{d}r} = \pm\left(1 - \frac{2GM}{r}\right)^{-1}. \tag{5.10}$$

This measures the slope of light cones on a spacetime diagram of the $t-r$ plane. Schwarzschild metric does not preserve light cone, as shown in figure 2.

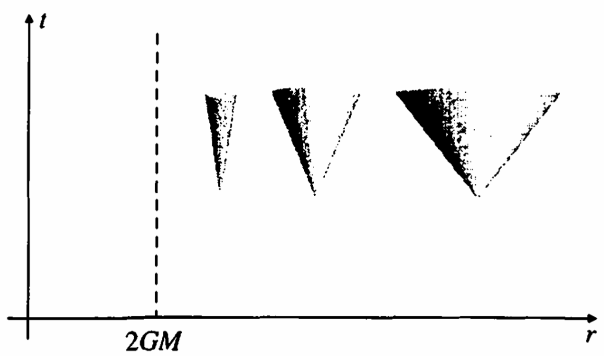

Figure 2: In Schwarzschild coordinates, light cones appear to close up as we approach $r = 2GM$.

We would never see the observer cross $r = 2GM$, we would just see them move more and more slowly (and become redder and redder).

- From Newton's theory, we are able to derive the escape velocity of a particle at distance r from a gravitating body of mass M:

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r}}. \tag{5.11}$$

If we naively ask where the Newtonian escape velocity equals the velocity of light, we find exactly $r = 2GM$. However, this is not the right way, though the results are coincidentally the same.

5.7 The Maximally Extended Schwarzschild Solution

- The **Kruskal diagram** explicitly shows geometric structure of Schwarzschild black hole, in which the diagram is divided into four regions. We say that the diagram represents the **maximal extension** of Schwarzschild geometry.
- The region I corresponds to $r > 2GM$, where our original coordinates are well-defined; The region II is the black hole. Nothing traveling region I to II returns; Region III represents a **white hole**, the time-reverse of region II, which is never observed; Region IV can be thought of as being connected to region I by a **wormhole**.

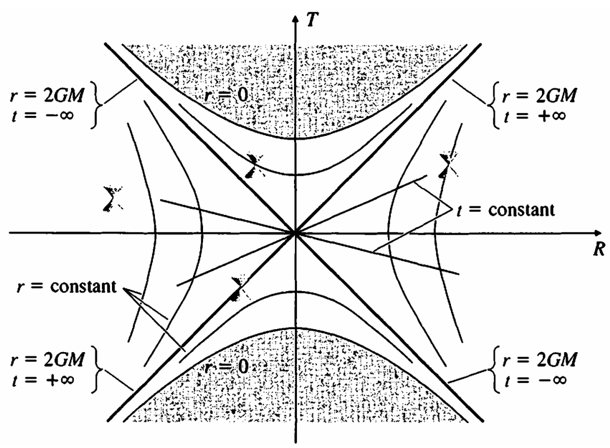

Figure 3: The Kruskal diagram.

5.8 Stars and Black Holes

- We model the star itself as a perfect fluid with energy-momentum tensor

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + p g_{\mu\nu}. \tag{5.12}$$

With a bunch of calculations, we obtain **Tolman-Oppenheimer-Volkoff equation**:

$$\frac{\mathrm{d}p}{\mathrm{d}r} = \frac{(\rho + p)\left[Gm(r) + 4\pi G r^3 p\right]}{r\left[r - 2Gm(r)\right]}. \tag{5.13}$$

This equation relates $p(r)$ to $\rho(r)$.

- **Buchdahl's theorem** states that any reasonable static, spherically symmetric interior solution has $M < 4R/9G$.

- The collapse of a star may eventually halted by **Fermi degeneracy pressure**. A stellar remnant supported by electron degeneracy pressure is called a **white dwarf**; a typical white dwarf is comparable in size to the earth.
- If the total mass if sufficiently high, the star will reach **Chandrasekhar limit**, where even the electron degeneracy pressure is not enough to resist the pull of gravity. When it is reached, the star is forced to collapse to an even smaller radius. At this point electrons combine with protons to make neutrons and neutrinos, and the neutrinos simply fly away. The result is a **neutron star**. The course [1] discusses these topics deeply.

6 More General Black Holes

6.1 The Black Hole Zoo

- **No hair theorem**: Stationary, asymptotically flat black hole solutions to general relativity coupled to electromagnetism that are nonsingular outside the event horizon are fully characterized by parameters of mass electric and magnetic charge, and angular momentum.

We can take a very complicated collection of matter, collapse it into a black hole, and end up with a configuration described completely by mass, charge, and spin.

- Taking quantum field theory into account, we inevitably meet with **information paradox**.
- In this chapter, we will discuss specific solutions corresponding to charged (Reissner-Nordström) and spinning (Kerr) black holes. As you will see, things are getting complicated, but quite interesting.

6.2 Event Horizons

- An event horizon is a **hypersurface** separating those spacetime that are connected to infinity by a timelike path from those that are not. For the Schwarzschild solution, the event horizon is a place where the light cones tilt over so that $r = 2GM$ is a null surface rather than a timelike surface.
- The Hawking-Penrose theorem demonstrates that once collapse reaches a certain point, evolution to a singularity is inevitable.
- Note that the concept of event horizon depends on the certain observer. There is just no need for us to specify the observer for Schwarzschild spacetime since its horizon is completely determined by intrinsic geometry.
- **Singularity theorem**: Let M be a manifold with a generic metric $g_{\mu\nu}$, satisfying Einstein's equation with the strong energy condition imposed. If there is a trapped surface in M, there must be either a closed timelike curve or a singularity.

To be precise, the singularity theorem states the idea that as long as GR is correct and causal structures are not violated, there must be a singularity in a universe with matter.

The GR itself is a theory at temperature of absolute zero. Time reversal is likely to violate the second law of thermodynamics. The third law of thermodynamics may prevent the formation of singularity. Among properties, merely "heat" and "gravity" are universal. The two concepts cannot be divided up as they are closely related. For more detailed discussions, refer to [2], which is enlightening.

- **Cosmic censorship conjecture**: Naked singularities cannot form in gravitational collapse from generic, initially nonsingular states in an asymptotically flat spacetime obeying the dominant energy condition. Note that the conjecture refers to the formation of naked singularities, not their existence.

6.3 Killing Horizons

6.4 Mass, Charge, and Spin

- How to define energy in curved spacetime remains a subtle problem. Taking electromagnetic field into considerations, we have **Komar integral**:

$$E_R = \frac{1}{4\pi G} \int_{\partial E} \mathrm{d}^2 x \sqrt{\gamma^{(2)}} n_\mu \sigma_\mu \nabla^\mu K^\nu , \tag{6.1}$$

which describes a conserved charge generated by a Killing vector. For example, the energy of a Schwarzschild black hole is

$$E_R = \frac{1}{4\pi G} \int \mathrm{d}\theta \, \mathrm{d}\phi \, r^2 \sin\theta \left(\frac{GM}{r^2} \right) . \tag{6.2}$$

This convinces us that we are on the right track. The quantity E_R is formally called **ADM energy**, rather than the ordinary energy we use everyday. We have the similar integral for angular momentum J.

- Yau and Schoen first proved the **positive energy theorem**[3]: The ADM energy of a nonsingular, asymptotically flat spacetime obeying Einstein's equation and dominant energy condition is non-negative. Furthermore, Minkowski is the only such spacetime with vanishing ADM energy. Edward Witten proved the theorem later, in a more physical way[4]. As one of the most talented theoretical physicists, he was awarded Fields Medal in 1990, for this outstanding work[5].

6.5 Charged (Reissner-Nordström) Black Holes

- **Reissner-Nordström metric**:

$$\mathrm{d}s^2 = -\Delta \, \mathrm{d}t^2 + \Delta^{-1} \, \mathrm{d}r^2 + r^2 \, \mathrm{d}\Omega^2 , \tag{6.3}$$

 where

$$\Delta = 1 - \frac{2GM}{r} + \frac{G(Q^2 + P^2)}{r^2} . \tag{6.4}$$

 In this expression, Q is the total electric charge, and P is the total magnetic charge. Monopoles are never observed in nature, but that won't bother us. The sign of Δ has physical significance, as listed below.

- $GM^2 < Q^2 + P^2$: In this case the coefficient Δ is always positive. this condition states that the total energy of the hole is less than the contribution from the electromagnetic field alone. This solution is generally considered to be unphysical.
- $GM^2 > Q^2 + P^2$: The metric has coordinate singularities at both r_+ and r_-. In both cases these could be removed by a change of coordinates as we did with Schwarzschild.
- $GM^2 = Q^2 + P^2$: This case is know as the extreme Reissner-Nordström solution where the mass is in some sense balanced by the charge. More specifically, two black holes with same-sign charges will attract each other gravitationally, but repel each other electromagnetically, and it turns out that these effects precisely cancel. In supersymmetric theories, extremal black holes can leave certain symmetries unbroken.

6.6 Rotating (Kerr) Black Holes

- **Kerr metric**:

$$
\begin{aligned}
\mathrm{d}s^2 = &-\left(1 - \frac{2GMr}{\rho^2}\right)\mathrm{d}t^2 - \frac{2GMar\sin^2\theta}{\rho^2}(\mathrm{d}t\,\mathrm{d}\phi + \mathrm{d}\phi\,\mathrm{d}t) \\
&+ \frac{\rho^2}{\Delta} + \rho^2\,\mathrm{d}\theta^2 + \frac{\sin^2\theta}{\rho^2}\left[(r^2+a^2)^2 - a^2\Delta\sin^2\theta\right]\mathrm{d}\phi^2 ,
\end{aligned}
\tag{6.5}
$$

  where

$$
\Delta(r) = r^2 - 2GMr + a^2, \tag{6.6}
$$

  and

$$
\rho^2(r,\theta) = r^2 + a^2\cos^2\theta. \tag{6.7}
$$

To verify that the mass M is equal to the Komar energy (6.1) is straightforward but tedious.

- As the case of Reissner-Nordström, we discuss three possibilities: $GM > a$, $GM = a$ and $GM < a$. The last case features a naked singularity, and the extremal case $GM = a$ is unstable, we will concentrate on $GM > a$. Then there are two radii at which Δ vanishes, given by

$$
r_\pm = GM \pm \sqrt{G^2M^2 - a^2}. \tag{6.8}
$$

  Both radii are null surfaces that will turn out to be event horizons.

- In fact, there is an ergosphere that wraps horizons around. It is possible to enter it and leave again, but you must move in the direction of the rotation of the black hole as if it is dragging you.

6.7 The Penrose Process and Black Hole Thermodynamics

- Particles' energy can be negative in the ergosphere, or accelerated until its energy is positive if it is to escape. The size of the ergosphere, the distance between the ergosurface and the event horizon, is not necessarily proportional to the radius of the event horizon, but rather to the black hole's gravity and its angular momentum.
- We can extract energy from a rotating black hole, this process is known as **Penrose process**. From calculation, one can extract energy no more than 29% of the black hole's original mass by Penrose process.
- Once a cup of coffee is tossed into the black hole, the second law of thermodynamics is violated, as the total entropy decreases. The no-hair theorem can no longer be sound to describe black hole. Hawking showed that quantum fields in a black hole background allow the hole to radiate at a temperature of $T = \kappa/2\pi$. Bekenstein has proposed a **generalized second law**, that the combined entropy of matter and black holes never decreases:

$$
\delta\left(S + \frac{A}{4G}\right) \geq 0. \tag{6.9}
$$

  This inequality is valid when black hole absorbs matter.

The black hole thermodynamics reveals so many new insights for quantum gravity that it is highly concerned these years. For more detailed contents on black hole thermodynamics, read the excellent notes (as a wonderful 2024 Christmas gift) [6] by Edward Witten, which provides concise interpretations. The lecture videos given at DESI are now available, on [7]. We will concentrate on this topic in Chapter 9.

The Playful Genius Behind the Cosmos

Roger Penrose, born on August 8, 1931, in Colchester, England, is a mathematical physicist and philosopher renowned for his profound contributions to science and thought. Growing up in an intellectually rich household, Penrose pursued mathematics at University College London and earned a Ph.D. from Cambridge. He revolutionized cosmology with the Penrose-Hawking singularity theorems, discovered aperiodic tilings that inspired quasicrystals, and developed Twistor Theory to bridge quantum mechanics and general relativity. A thinker who spans disciplines, Penrose's ideas on consciousness and artificial intelligence, expressed in works like *The Emperor's New Mind*, have provoked widespread debate and exploration.

Penrose's life is filled with delightful stories that reflect his inventive mind and penchant for exploring the unconventional. One of his most famous creations, the Penrose stairs, was born from his fascination with optical illusions and impossible figures. This idea later inspired M.C. Escher's famous lithograph *Ascending and Descending*, showcasing the productive interplay between art and science. Penrose humbly recounted that he initially sketched the stairs while idly doodling with his father, unaware it would become an iconic symbol of mathematical artistry.

As a student, Penrose's unconventional thinking often left his teachers baffled. During a geometry exam, he devised a solution so unorthodox that it appeared incorrect at first glance. After careful review, the examiners realized his method was not only valid but brilliantly innovative. This anecdote encapsulates his lifelong tendency to see the world through a unique lens.

Another whimsical moment came during his presentation on Twistor Theory, a bold attempt to unify quantum mechanics and general relativity. An audience member skeptically asked, "But does it work?" With characteristic wit, Penrose responded, "Ah, that's an entirely different question!" This humility and humor endear him to colleagues and students alike.

Penrose's relationship with Stephen Hawking also yields charming stories. The two collaborated on singularity theorems, sparking a lifelong intellectual camaraderie. Penrose fondly recalls how their friendly debates often evolved into playful arguments, with Hawking once declaring, "Roger, you're the smartest person I know, but you're completely wrong about this!" Despite their disagreements, their mutual respect and shared curiosity fueled some of the most significant advances in cosmology.

Penrose's curiosity extends beyond academia. An avid doodler, he frequently sketches intricate diagrams that capture his ideas visually. These sketches often appear in his lectures and books, providing a window into his thought process. His interest in artistic representation also connects to his exploration of tiling patterns. While developing Penrose tilings, he reportedly experimented with colorful cutouts and mosaics on his living room floor, driven by a childlike fascination with patterns.

Perhaps one of the most endearing stories about Penrose is his view on failure. When reflecting on moments when his theories did not pan out, he once said, "It's not about being right or wrong; it's about the joy of discovering something new." This spirit of curiosity and resilience defines his work and inspires generations of scientists to embrace the unknown with wonder.

Roger Penrose's anecdotes illuminate his genius and humanity, painting a portrait of a scientist whose playfulness and creativity are as boundless as the cosmos he seeks to understand.

7 Perturbation Theory and Gravitational Radiation

7.1 Linearized Gravity and Gauge Transformations

- We can decompose the metric into the flat Minkowski metric plus a small perturbation

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \tag{7.1}$$

where $h_{\mu\nu} \ll 1$. We can think of the linearized version of general relativity as describing a theory of a symmetric tensor field $h_{\mu\nu}$ propagating on a flat background spacetime.

- The formula

$$h^{(\epsilon)}_{\mu\nu} = h_{\mu\nu} + 2\epsilon\partial_{(\mu}\xi_{\nu)} \tag{7.2}$$

represents the change of the mtric perturbation under an infinitesimal diffeomorphism along the vector field $\epsilon\xi^{\mu}$: we will call this a **gauge transformation** in linearized theory. The transformation leaves the curvature unchanged.

7.2 Degrees of Freedom

- According to symmetry of perturbation term $h_{\mu\nu}$, we write it as

$$\begin{aligned} h_{00} &= -2\Phi \\ h_{0i} &= w_i \\ h_{ij} &= 2s_{ij} - 2\Psi\delta_{ij}, \end{aligned} \tag{7.3}$$

where Ψ encodes the trace of h_{ij} and s_{ij} is traceless. The entire metric is now written as

$$ds^2 = -(1+2\Phi)\mathrm{d}t^2 + w_i(\mathrm{d}t\mathrm{d}x^i + \mathrm{d}x^i\mathrm{d}t) + [(1-2\Psi)\delta_{ij} + 2s_{ij}]\mathrm{d}x^i\mathrm{d}x^j. \tag{7.4}$$

s_{ij} is known as the **strain**.

7.3 Newtonian Fields and Photon Trajectories

7.4 Gravitational Wave Solutions

7.5 Production of Gravitational Waves

7.6 Energy Loss Due to Gravitation Radiation

7.7 Detection of Gravitational Waves

- The gravitational wave was first detected in 2015, by the Laser Interferometer Gravitational-Wave Observatory (LIGO)[8]. The observed gravitational waves originate from the merger of two black holes with a combined mass of approximately 65 times that of the Sun. Due to the tiny changes caused by gravitational wave, the observatory is actually a giant Michelson interferometer, with each of its arm constructed to be about 4 kilometers long.
- For typical parameters we can take both black holes to be 10 solar mass, the binary to be at cosmological distances about 100 Mpc, and the components to be separated by ten times their

Schwarzschild radii:

$$\begin{aligned} R_S &\sim 10^6 \text{cm} \\ R_S &\sim 10^7 \text{cm} \\ r &\sim 10^{26} \text{cm}. \end{aligned} \tag{7.5}$$

Such a source is thus characterized by

$$f \sim 10^2 \text{s}^{-1}, \quad h \sim 10^{-21}. \tag{7.6}$$

- The physical effect of a passing gravitational wave is to slightly perturb the relative positions of free-falling masses. If two test masses are separated by a distance L, the change in their distance will be roughly
$$\frac{\delta L}{L} \sim h. \tag{7.7}$$
- In the interferometer, during 100 round trips through the cavity arms, the accumulated phase shift will be
$$\delta\phi \sim 200 \left(\frac{2\pi}{\lambda}\right) \delta L \sim 10^{-9}. \tag{7.8}$$

8 Cosmology

8.1 Maximally Symmetric Universe

- The universe is **homogeneous** and **isotropic** on large scales. In other words, the universe is maximally symmetric. As we will see, these conditions bring about problems. In such a spacetime we have (recall equation (3.24))

$$R_{\rho\sigma\mu\nu} = \kappa(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}). \tag{8.1}$$

  By taking the trace of Riemann tensor and specifying to four dimensions, calculation gives

$$R_{\mu\nu} = 3\kappa g_{\mu\nu}, \quad R = 12\kappa. \tag{8.2}$$

  Furthermore we find that the energy-momentum tensor is proportional to the metric.

- The maximally symmetric spacetime with vanishing curvature is the Minkowski space, with metric

$$ds^2 = -\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\,; \tag{8.3}$$

  The maximally symmetric spacetime with positive curvature ($\kappa > 0$) is the **de Sitter space** (dS). De Sitter space describes a spatial three-sphere that initially shrinks, reaching a minimum size at $t = 0$, and then re-expands. The topology of de Sitter is $\mathbb{R} \times S^3$. This means that every point on the time axis corresponds a separate 3-sphere.

- Another coordinate choice gives the **Einstein static universe**, described by the metric

$$ds^2 = -\mathrm{d}t^2 + \mathrm{d}\chi^2 + \sin^2\chi\,\mathrm{d}\Omega_2^2\,. \tag{8.4}$$

- A similar hyperboloid construction reveals the $\kappa < 0$ spacetime of maximal symmetry, known as **anti-de Sitter space** (AdS). The property that there is no closed timelike curve in this space can be taken as the definition.

The anti-de Sitter space plays an important role in research of high-energy formal theory. In 1997, a stunning article was delivered by Juan Maldacena, which stated the correspondence (or equivalence) between quantum gravity in $(d+1)$- dimensional AdS space and conformal field theory at the boundary of d-dimensional spacetime[9]. For a comprehensive review article, see [10][11]. The de Sitter space is counter-intuitively rather hard to deal with, read [12].

- The maximally symmetric spacetimes (Minkowski, de Sitter and anti-de Sitter) may be thought of as ground states of GR. Because of the existence of matter, none of them are reasonable models of the real world.

8.2 Robertson-Walker Metrics

- To describe the real world, we tend to posit that the universe is spatially homogeneous and isotropic, but evolving in time. We take the metric

$$ds^2 = \mathrm{d}t^2 + R^2(t)d\sigma^2, \tag{8.5}$$

  where $R(t)$ is a function known as **scale factor**. It has the units of distance.

- Only the **comoving** observer will think that the universe looks isotropic.

- **Robertson-Walker metric**:

$$ds^2 = -\,\mathrm{d}t^2 + R^2(t)\left(\frac{\mathrm{d}r^2}{1-kr^2} + r^2 d\Omega^2\right). \tag{8.6}$$

  The Ricci scalar of this is

$$R = 6\left[\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2}\right]. \tag{8.7}$$

- The value of k corresponds to different cases: $k = 0$ for flat spacetime; $k = 1$ for **closed** spacetime; $k = -1$ for **open** spacetime.

8.3 The Friedmann Equation

- Combining Robertson-Walker metric and Einstein equation together, we obtain the **Friedmann equation**:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{\kappa}{a^2}. \tag{8.8}$$

  If we know the dependence of ρ on a, it is enough to solve for $a(t)$. The second Friedmann equation is written as

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p). \tag{8.9}$$

- The two most popular examples of cosmological fluids are known as **matter** and **radiation**. Matter is also know as *dust*. A matter-dominated universe has

$$\rho_M \propto a^{-3}. \tag{8.10}$$

  This is simply interpreted as the decrease in the number density of particles as the universe expands. A matter-dominated universe has

$$\rho_R \propto a^{-4}; \tag{8.11}$$

  A vacuum-dominated universe has

$$\rho_M \propto a^0. \tag{8.12}$$

  The de-Sitter space and anti-de Sitter space are both vacuum-dominated solutions.

- The rate of expansion is characterized by the **Hubble parameter**

$$H = \frac{\dot{a}}{a}. \tag{8.13}$$

  The value of the Hubble parameter at the present epoch is the Hubble constant H_0. Current measurements lead us to believe that the Hubble constant is 70 ± 10km/sec/Mpc. According to astronomical observation and basic knowledge, the age of the universe is about 13.8 billion years.

- The **deceleration parameter**

$$q = -\frac{a\ddot{a}}{\dot{a}^2}, \tag{8.14}$$

  which measures the rate of change of the rate of expansion.

Cosmology is in some sense closely related to particle physics in recent years. Lectures given by Zhongzhi Xianyu at Yau Center of Southeast University provided comprehensive knowledge about this[13].

8.4 Evolution of the Scale Factor

8.5 Redshifts and Distances

- The particle "slows down" with respect to the comoving coordinates as the universe expands.
- The **redshift** of an object tells us the scale factor when the photon was emitted:

$$z_{\text{em}} = \frac{\lambda_{\text{obs}} - \lambda_{\text{em}}}{\lambda_{\text{obs}}}. \tag{8.15}$$

  This is not the same as the conventional Doppler effect; it is the expansion of the space, *not* the relative velocities of the observer and emitter.

- In 1929, American astronomer Hubble discovered that the farther galaxies recess faster. This is thus called **Hubble law**:

$$v = H_0 d_p, \tag{8.16}$$

  which states that the observed recession velocity is directly proportional to the distance, for galaxies that are not too far away. To understand the law better, imagine two person pulling a rope from the opposite directions.

- The **luminosity distance** is defined to satisfy

$$d_L^2 = \frac{L}{4\pi F}, \tag{8.17}$$

  where L is the absolute luminosity of the source and F is the flux measured by the observer.

8.6 Gravitational Lensing

8.7 Our Universe

- A simple candidate for a dynamical source of dark energy is provided by a slowly-rolling scalar field. Consider a field ϕ with the usual action

$$S = \int \mathrm{d}^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) \right]. \tag{8.18}$$

Under some plausible assumptions, the equation of motion is

$$\ddot{\phi} + 3H\dot{\phi} + \frac{\mathrm{d}V}{\mathrm{d}\phi} = 0. \tag{8.19}$$

We see that the Hubble parameter acts as a friction term; the field will tend to toll down the potential, but when H is too large the motion will be damped.

8.8 Inflation

- The most straight forward way is to use the vacuum energy provided by the potential of a scalar field, the **inflaton**. We have the equation of motion for a scalar field in an RW metric

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0, \tag{8.20}$$

  as well as the Friedmann equation

$$H^2 = \frac{1}{3m_p} \left(\frac{1}{2}\dot{\phi} + V(\phi) \right). \tag{8.21}$$

- The first problem bothering us is **flatness problem**. According to Friedmann equation, the curvature of universe increase as universe evolves. However, the universe we observed is nearly flat.
- **The horizon problem** arises from the fact that distinct patches of the CMB sky were causally disconnected (approximately 40000 independent regions[14]), however, the temperature differences between patches are tiny (about 10^{-5})). This motivated us to propose the **inflation** theory.
- The theory looks trivial if the potential energy of the scalar field remains constant, thus we have to introduce **Slow-roll parameters**. They are defined as

$$\epsilon = \frac{1}{2} m_p^2 \left(\frac{V'}{V} \right)^2 , \tag{8.22}$$

$$\eta = m_p^2 \left(\frac{V''}{V} \right) . \tag{8.23}$$

The inflation is driven when these two parameters are much smaller than 1. This is **slow-roll condition**.

9 Quantum Field Theory in Curved Spacetime

9.1 Introduction

- Black holes are not really black, but instead emit thermal radiation at a **Hawking temperature** proportional to the surface gravity:
$$T = \frac{\kappa}{2\pi}. \tag{9.1}$$
- The familiarity with QFT in flat spacetime is by no means necessary for studying QFT in curved spacetime. The only prerequisite is a familiarity with the basics of ordinary quantum mechanics.

9.2 Quantum Mechanics

Quantum mechanics is profoundly different from classical mechanics, it is of significance to impose the basic assumptions:

(1) The state of the system is represented as an element of **Hilbert space**. Mathematically, a Hilbert space is a complex vector space equipped with inner product. We denote the elements of Hilbert space as $|\psi\rangle$ and elements of dual space as $\langle\psi|$, their inner product obeys
$$\langle\psi_2|\psi_1\rangle^* = \langle\psi_1|\psi_2\rangle. \tag{9.2}$$

(2) Observables are represented as **self-adjoint operators** on the Hilbert space. In QM we generally call them **Hermitian** operator,
$$A^\dagger = A, \tag{9.3}$$
where A obeys
$$\langle\psi_2|A\psi_1\rangle = \langle A^\dagger\psi_2|\psi_1\rangle. \tag{9.4}$$

(3) Evolution of the system may be represented in one of two ways: as unitary evolution of the state vector in Hilbert space (the **Schrödinger picture**), or by keeping the state fixed and allowing the observables to evolve according to equations of motion (the **Heisenberg picture**).

- The equation of motion of a harmonic oscillator is **Schrödinger's equation**
$$i\partial_t\psi = H\psi. \tag{9.5}$$
The solution to the equation is
$$\psi_n(x,t) = e^{-\omega x^2/2}H_n(\sqrt{\omega}x)e^{-iE_nt}, \tag{9.6}$$
and
$$E_n = \left(n+\frac{1}{2}\right)\omega, \tag{9.7}$$
they are all eigenfunctions of H. E_n is the energy eigenvalue, it is evident that the system possesses a zero-point energy of $E_0 = \omega/2$.
- Using creation and annihilation operator, the Hamiltonian can be expressed as
$$H = \left(\hat{a}^\dagger\hat{a}+\frac{1}{2}\right)\omega. \tag{9.8}$$
- Given Schrödinger's equation, any state can be written formally as some fixed initial state acted

on by a unitary time-evolution operator

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,, \tag{9.9}$$

where

$$U(t) = \exp\left(-i\int H\,\mathrm{d}t\right) \tag{9.10}$$

satisfies Schrödinger's equation as well.

The mathematical structure of quantum mechanics is rather complete. A researcher with mathematical background may concentrate more on functional analysis or operator algebra. If you are interested, refer to [15][16].

9.3 Quantum Field Theory in Flat Spacetime

- The Klein-Gordon Lagrangian is given by

$$\mathscr{L} = -\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}m^2\phi^2, \tag{9.11}$$

  for which we can easily write down the plane-wave solution as:

$$\phi(x^\mu) = \phi_0 e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}. \tag{9.12}$$

In quantum field theory, particles are viewed as energy eigenstates of definite momenta, our modes are plane waves that extent throughout the space. However, this no longer holds for curved spacetime.

9.4 Quantum Field Theory in Curved Spacetime

- By promoting the ordinary derivative to covariant derivative and introducing the general spacetime metric, we can easily write down the Lagrangian density of non-interacting massive scalar field in curved spacetime:

$$\mathscr{L} = \sqrt{-g}\left(-\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - \frac{1}{2}m^2\phi^2 - \xi R\phi^2\right). \tag{9.13}$$

  Minimal coupling sets $\xi = 0$ and **conformal coupling** sets $\xi = 1/6$ in four dimensions, though we have no preference on the choice of ξ.

- Following the standard procedure of canonical quantization, this gives equation of motion

$$\Box\phi - m^2\phi - \xi R\phi = 0, \tag{9.14}$$

  and a nontrivial commutation relation

$$[\phi(t,\vec{x}), \pi(t,\vec{x}')] = \frac{i}{\sqrt{-g}}\delta^{(n-1)}(\vec{x}-\vec{x}'). \tag{9.15}$$

- We again use positive- and negative-frequency modes to expand our field as

$$\phi = \sum_i\left(a_i f_i + a_i^\dagger f_i^*\right), \tag{9.16}$$

where a_i and $a_i^\dagger$ satisfy commutation relation

$$\left[a_i, a_j^\dagger\right] = \delta_{ij}. \tag{9.17}$$

- We can declare the excitations created by $a_i^\dagger$ to be particles and construct Fock space, nevertheless the basis modes f_i are highly nonunique. Another observer may use b_i and $b_i^\dagger$ as his creation and annihilation operator. The two observers may disagree on how many particles are observed. To see this, it is convenient to expand each set of modes in terms of the other:

$$\begin{aligned} g_i &= \sum_j \left(\alpha_{ij} f_j + \beta_{ij} f_j^*\right) \\ f_i &= \sum_j \left(\alpha_{ji} g_j - \beta_{ji} f_g^*\right). \end{aligned} \tag{9.18}$$

- The transformation from one set of basis modes into another is known as a **Bogoliubov transformation**[1]. Matrices α_{ij} and β_{ij} implementing the transformation are **Bogoliubov coefficients**, which satisfies their own normalization conditions.
- The number of g-particles in the f-vacuum can be expressed in terms of the Bogoliubov coefficients as

$$\left\langle 0_f | n_{gi} | 0_f \right\rangle = \sum_j |\beta_{ij}|^2. \tag{9.19}$$

  If any of the β_{ij} are nonvanishing, the vacuum states will not coincide. The β_{ij} describes the admixture of creation operators from one basis into the annihilation operators in the other basis.
- We see that QFT in curved spacetime shares most of the basic features of QFT in flat spacetime; the crucial difference involves what we cannot do, namely decide on a natural set of basis modes that all inertial observers would identify as particles.

The books [17][18] are well-known textbooks.

9.5 The Unruh Effect

- The **Unruh effect** states that a uniformly accelerating observer in the traditional Minkowski vacuum state will observe a thermal spectrum of particles. It is a manifestation of the idea that observers with different notions of positive- and negative-frequency modes will disagree on the particle content of a given state.
- The region with the metric

$$ds^2 = e^{2a\xi}(-\mathrm{d}\eta^2 + \mathrm{d}\xi^2) \tag{9.20}$$

  is known as the **Rindler space**. A Rindler observer is one moving along a constant-acceleration path, whose trajectory corresponds to a hyperbola in spacetime coordinates.
- The temperature $T = a/2\pi$ is what would be measured by an observer moving along the path $\xi = 0$, which feels an acceleration $a = \alpha$. The Unruh effect tells us that an accelerated observer will detect particles in the Minkowski vacuum. The effect is so hard to detect that 1K of temperature elevation requires an acceleration to be approximately $10^{20}\mathrm{m/s^2}$.

[1]There are typos on these equations, on the page 398.

9.6 The Hawking Effect and Black Hole Evaporation

- When the wavelength of the emitted radiation is comparable to the Schwarzschild radius, the spectrum of emmited radiation takes the form

$$\langle \hat{n}_\omega \rangle = \frac{\Gamma(\omega)}{e^{2\pi(\omega-\mu)/\kappa} \pm 1}. \tag{9.21}$$

  Here $\Gamma(\omega)$ is a greybody factor, which can be thought of as arising from backscattering of wavepackets off of the gravitational field and into the black hole.

- The fact that black holes emit thermal radiation is surprising. Think about a virtual particle pair created near the horizon, occasionally one member of which will fall into the black hole while its partner escapes to infinity. It is these escaping particles that we observe as Hawking radiation.

- Considering the black hole thermodynamics, a single million-solar-mass black hole has more entropy than all of the matter in the visible universe.

- Article [19] provided interpretations and wonderful diagrams on this topic.

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A Maps between Manifolds

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